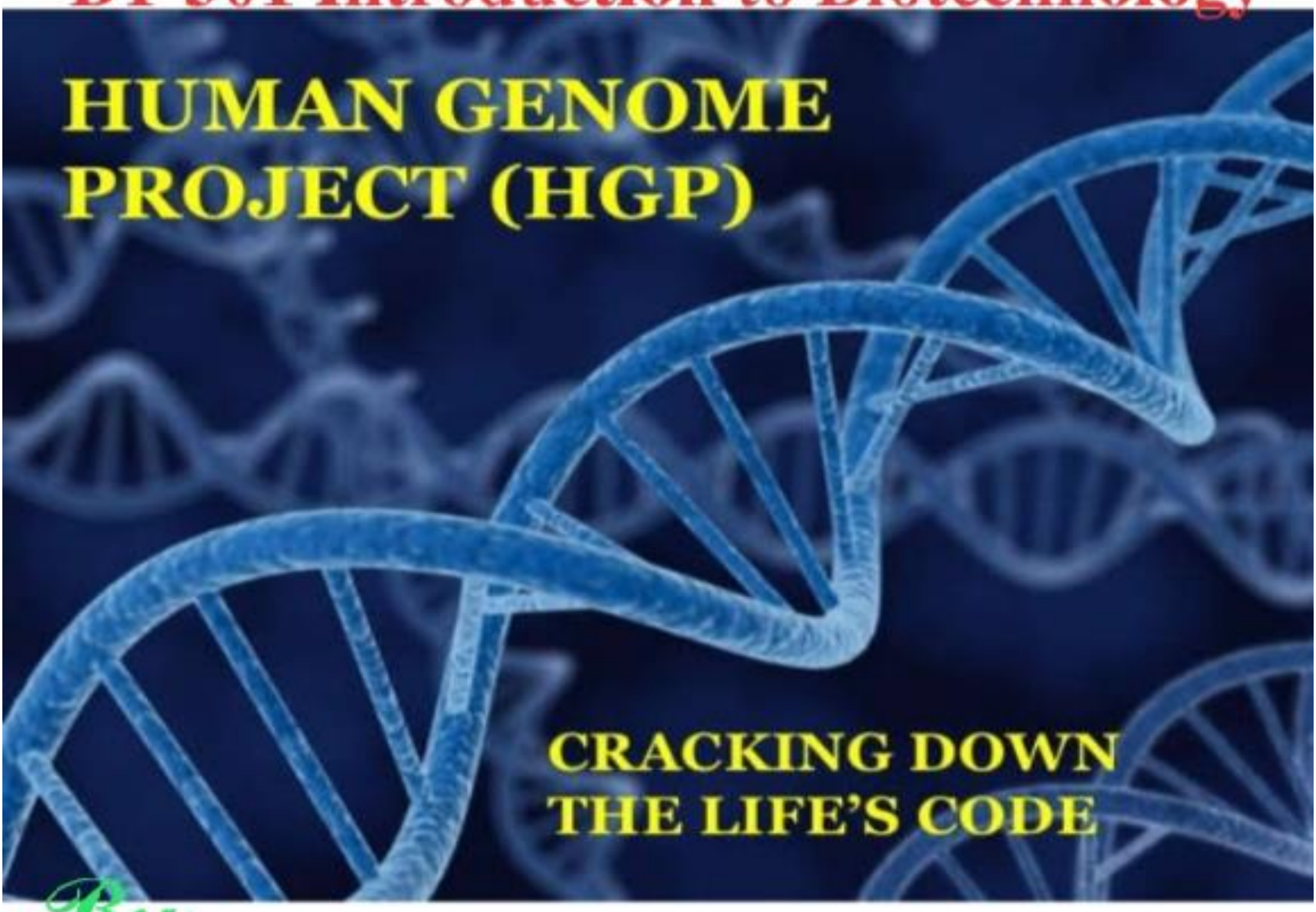
G D B

BT-301 Introduction to Biotechnology



By

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Solution:

HGP and disease risk prediction:

The Human Genome Project has made it easier to genetically map and clone mutant genes which predispose to a great many human diseases. In addition to diseases inherited in a simple Mendelian fashion, there are many degenerative and infectious diseases, and sensitivities to environmental insults, in which the genetic make-up of an individual contributes to the course of the disease. Once a gene or genes associated with disease has been cloned, it is possible to design DNA-based diagnostics to detect altered forms of the gene which predispose to disease. The ability to predict the development of disease makes possible early intervention to limit the severity of a disease or to use gene therapy to cure inherited disorders.

Example:

Breast cancer prediction:

Members of some families are particularly at risk of developing certain types of breast cancer, because they carry faulty alleles. Following the HGP, these alleles have been identified. It is now possible to test people to see if they have these alleles in their DNA. This will predict the likelihood of them developing the disease.

HGP and Personalized medicine:

Personalized medicine is an emerging practice of medicine that uses an individual's genetic profile to guide decisions made in regard to the prevention, diagnosis, and treatment of disease. Knowledge of a patient's genetic profile can help doctors select the proper medication or therapy and administer it using the proper dose or regimen. Personalized medicine is being advanced through data from the Human Genome Project.

HGP is playing a big role in the emergence of personalized medicine, 'cause it gives us a window in a very specific molecular way into those differences between us and allows the opportunity for making individual predictions about disease risk that can help somebody choose a prevention plan that is right for them. It also allows the possibility in some instances of picking the right drug at the right dose for the right person instead of the "one size fits all" approach to drug therapy. And ultimately it will be hard to see how any kind of medicine will not be affected by this as we learn more and more about the individual, and as many of us find our complete genomes being sequenced and placed into our medical records to empower that kind of personalized approach.

Example:

Targeted therapies – Cancer

A huge breakthrough in medicine has been the ability to sequence the DNA in cancer cells. The sequence can be compared to the sequence found by the Human Genome

Project. This allows scientists to work out which genes are mutated and this gives them ideas for developing medicines. For example, it was found that certain breast cancers have a mutation in a gene called HER-2. By finding this out, scientists were then able to produce a monoclonal antibody that was specifically targeted to cancer cells that had the mutation. The treatment is very successful and could not have been developed without our detailed knowledge of DNA sequencing and the human genome.

Regards:

BT-301 Expert Zoologists